



HUMAN BLOOD SUBSTITUTES

8. Human Blood Substitutes

- Human blood substitutes are **synthetic products** that are **designed to act** as a **replacement for blood** in the human body.
- **Basic Requirement for Human Blood Substitutes:**
 - Effective Oxygen Transport
 - Safety and Compatibility
 - Storage and Transport
 - Cost-Effectiveness and Scalability

Types of HBS

- There are two types of human blood substitutes –
 1. Hemoglobin-based oxygen carriers (HBOCs) and
 2. Perfluorocarbons (PFCs)
- 1. HBOCs: are based on the **hemoglobin molecule**, which is the protein in red blood cells that carries oxygen to the body's tissues.
- Hemoglobin is **extracted** from human or animal blood and **then modified** to create a **stable, synthetic version**.
- When introduced into the body, HBOCs can help to **increase the amount of oxygen available to the tissues**, which can be important in situations where the body is unable to produce or transport enough red blood cells.

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- They are **made by isolating hemoglobin**, the **protein responsible for carrying oxygen** in red blood cells, and formulating it into a **solution or suspension** that can be infused into a patient's bloodstream.

Advantages:

- Increased oxygen-carrying capacity
- Universal compatibility
- Longer shelf life
- Reduced risk of infections
- Availability in remote or challenging settings

Limitations/Risks:

- **Limited oxygen release** - inadequate oxygen delivery to tissues in certain conditions
- **Short half-life** - This means that the HBOCs may be rapidly cleared from circulation, reducing their effectiveness and requiring more frequent doses or infusions.
- **Nitric oxide scavenging** - Excessive nitric oxide scavenging by HBOCs can lead to vasoconstriction, impairing blood flow to vital organs and potentially causing adverse cardiovascular effects.



Examples of HBOCs

- **Hemopure:** Hemopure is an HBOC that is made from bovine hemoglobin. It has been approved for use in South Africa, Russia, and some other countries.
- **Oxyglobin:** Oxyglobin is another HBOC that is made from bovine hemoglobin. It is approved for veterinary use in the United States and has been used to treat anemia in dogs.

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- **Hemospan:** Hemospan is an HBOC that is being developed by Sangart Inc. It is currently in clinical trials and has shown promise in increasing oxygen delivery to tissues.
 - **MP4OX:** MP4OX is an HBOC that is being developed by Baxter Healthcare. It is designed to increase oxygen delivery to tissues and also to scavenge harmful free radicals in the bloodstream.
 - **Hemolink:** Hemolink is an HBOC that is being developed by Hemosol Inc. It is designed to be used in trauma and surgical settings and has shown promise in improving oxygen delivery to tissues.
 - **Note:** Many countries have not yet given regulatory approval for clinical usage of HBOCs

2. Perfluorocarbons (PFCs):

- They are synthetic molecules that are **similar in structure to the hemoglobin** molecule.
- However, unlike HBOCs, they **do not require modification from natural sources**.
- PFCs are **able to dissolve oxygen** and **transport it throughout the body**, similar to the way that red blood cells work.



Advantages of PFCs

- High oxygen-carrying capacity
- Improved oxygen solubility
- Stability and long shelf life
- No blood typing or cross-matching required
- Reduced risk of infection transmission
- Compatibility with diagnostic tests

Limitations of PFCs

- Limited oxygen offloading
- Need for specialized administration methods - such as emulsification or encapsulation.
- Short half-life
- Clearance and elimination - primarily eliminated from the body through the lungs.
- Side effects and toxicity
- Regulatory considerations

Examples of PFCs

- **Perftoran:** Perftoran is a PFC that was developed in Russia and is used in several countries, including Russia, Ukraine, and China. It has been used in the treatment of a variety of conditions, including trauma, heart attack, and stroke.
- **Oxycyte:** Oxycyte is a PFC that is being developed by Oxygen Biotherapeutics. It is currently in clinical trials and has shown promise in increasing oxygen delivery to tissues in patients with traumatic brain injury.

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- **Oxycyte PFC Emulsion:** This is another PFC-based blood substitute being developed by Oxygen Biotherapeutics. It is designed to be used as an oxygen carrier during surgery and other medical procedures.
 - **Hemopure-PFC:** Hemopure-PFC is a hybrid blood substitute that combines a PFC with a hemoglobin-based oxygen carrier. It is being developed by HbO₂ Therapeutics and has shown promise in increasing oxygen delivery to tissues in preclinical studies.
 - **Note:** It's important to note that while these technologies show promise, they are still in development and further studies are needed to evaluate their safety and effectiveness.
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